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Facultad de Psicología
UNIVERSIDAD DE LA REPÚBLICA

CIBPsi
CENTRO DE INVESTIGACIÓN BÁSICA EN PSICOLOGÍA

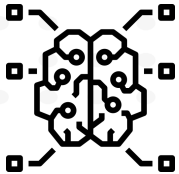
Modelos de lenguaje como herramientas y modelos neurocognitivos… y viceversa

Álvaro Cabana

Programa Ciencia de Datos y Psicometría
Instituto de Fundamentos y Métodos

UNIVERSIDAD DE LA REPÚBLICA URUGUAY

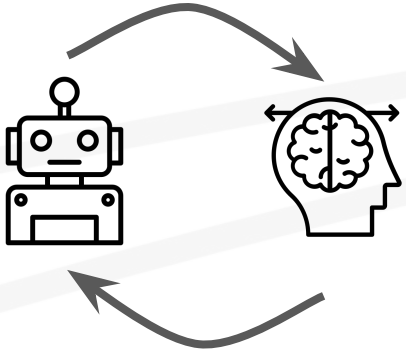
Redes neuronales artificiales



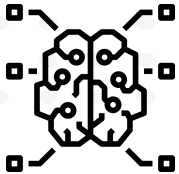
Construir para entender la mente

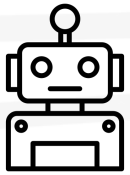
Modelos de redes neuronales

- Surgen para entender la mente
- Se usan para construir máquinas



Redes neuronales artificiales





- **1943** McCulloch y Pitts: primera neurona artificial digital
- **1958** Rosenblatt: perceptrones
- **1969** Minsky: libro que entierra a los perceptrones
- **1984** Rumelhart, Hinton, Williams: backpropagation y cómo entrenar tu perceptrón multicapa
- **1991** Elman: red recurrente simple

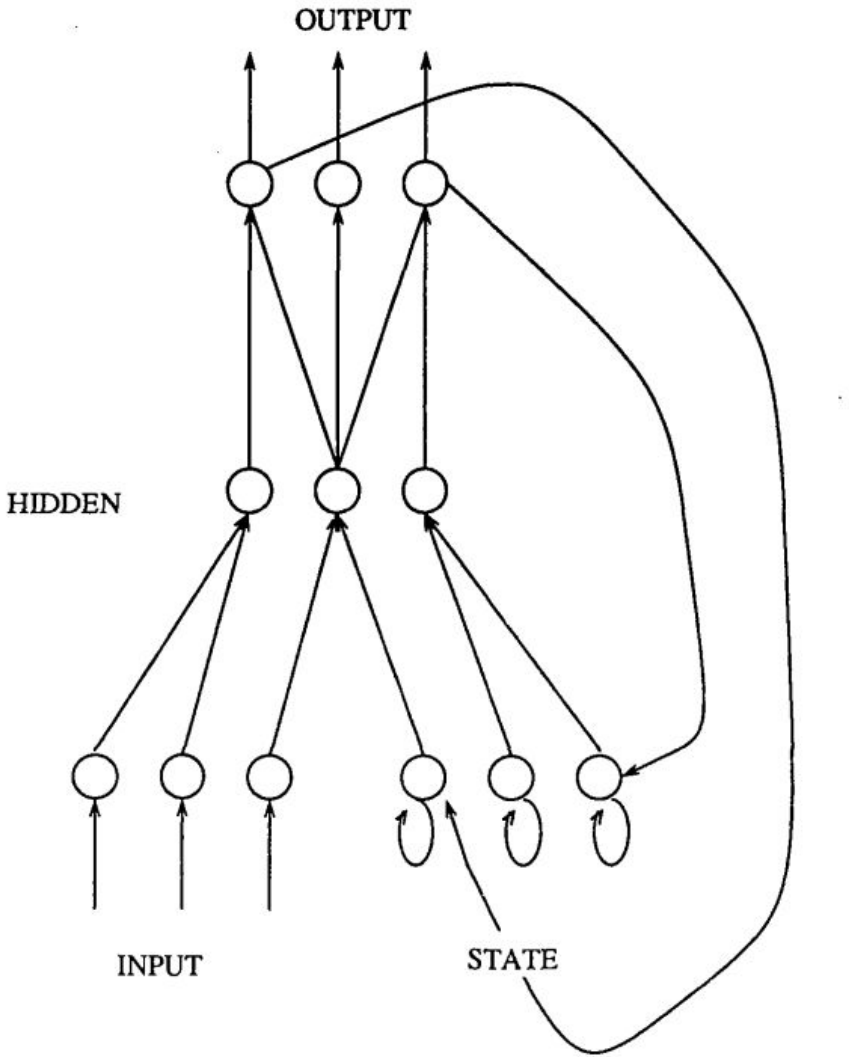
COGNITIVE SCIENCE **14,** 179–211 (1990)

Finding Structure in Time

JEF

Universit

Time underlies many interesting human behaviors. Thus, the question of how to represent time in connectionist models is very important. One approach is to represent time implicitly by its effects on processing rather than explicitly (as in a spatial representation). The current report develops a proposal along these lines first described by Jordan (1986) which involves the use of recurrent links in order to provide networks with a dynamic memory. In this approach, hidden unit patterns are fed back to themselves; the internal representations which develop thus reflect task demands in the context of prior internal states. A set of simulations is reported which range from relatively simple problems (temporal version of XOR) to discovering syntactic/semantic features for words. The networks are able to learn interesting internal representations which incorporate task demands with memory demands; indeed, in this approach the notion of memory is inextricably bound up with task processing. These representations reveal a rich structure, which allows them to be highly context-dependent, while also expressing generalizations across classes of items. These representations suggest a method for representing lexical categories and the type/token distinction.


OUTPUT
HIDDEN
INPUT
STATE

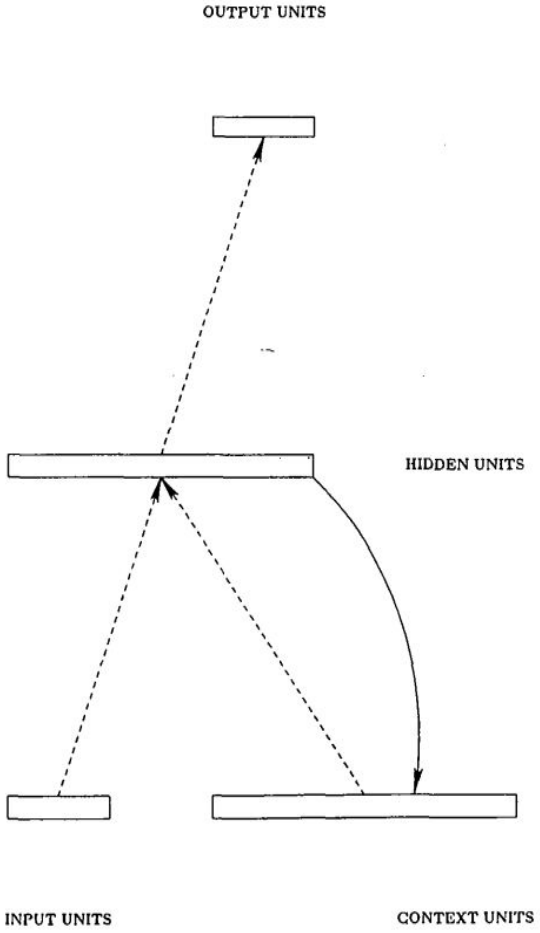

OUTPUT UNITS
HIDDEN UNITS
INPUT UNITS
CONTEXT UNITS

TABLE 4
Templates for Sentence Generator

WORD 1	WORD 2	WORD 3
NOUN-HUM	VERB-EAT	NOUN-FOOD
NOUN-HUM	VERB-PERCEPT	NOUN-INANIM
NOUN-HUM	VERB-DESTROY	NOUN-FRAG
NOUN-HUM	VERB-INTRAN	
NOUN-HUM	VERB-TRAN	NOUN-HUM
NOUN-HUM	VERB-AGPAT	NOUN-INANIM
NOUN-HUM	VERB-AGPAT	
NOUN-ANIM	VERB-EAT	NOUN-FOOD
NOUN-ANIM	VERB-TRAN	NOUN-ANIM
NOUN-ANIM	VERB-AGPAT	NOUN-INANIM
NOUN-ANIM	VERB-AGPAT	
NOUN-INANIM	VERB-AGPAT	
NOUN-AGRESS	VERB-DESTROY	NOUN-FRAG
NOUN-AGRESS	VERB-EAT	NOUN-HUM
NOUN-AGRESS	VERB-EAT	NOUN-ANIM
NOUN-AGRESS	VERB-EAT	NOUN-FOOD

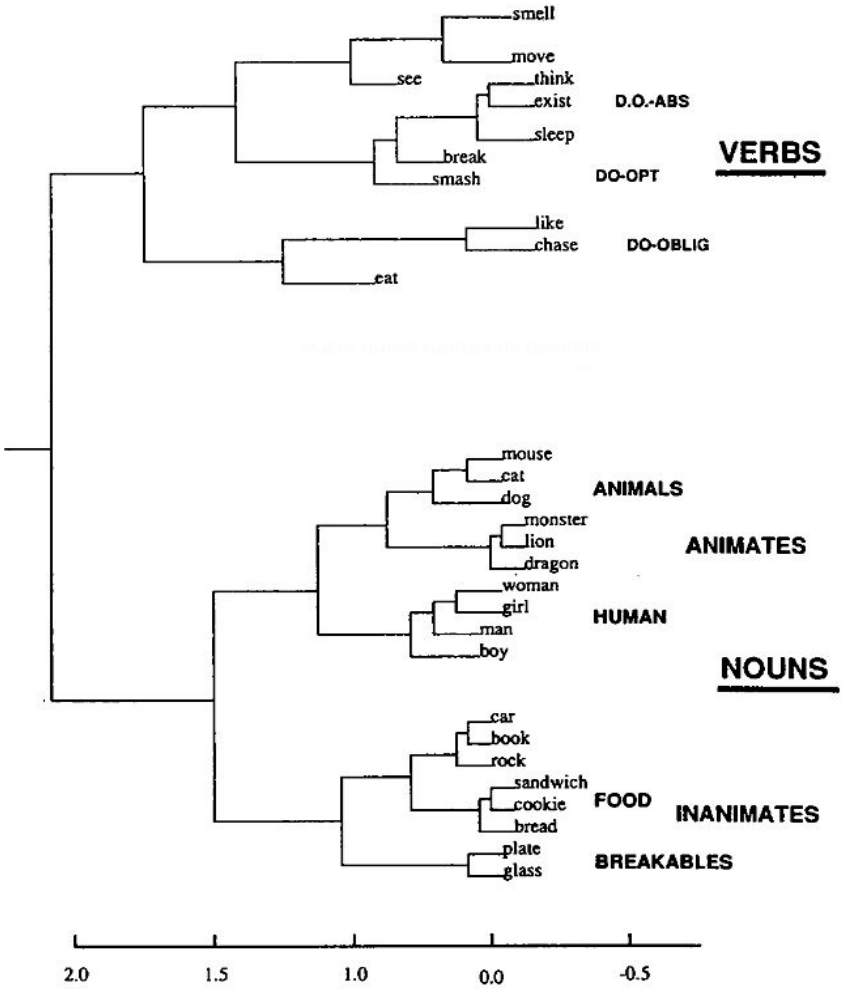
TABLE 3
Categories of Lexical Items Used in Sentence Simulation

Category	Examples
NOUN-HUM	man, woman
NOUN-ANIM	cat, mouse
NOUN-INANIM	book, rock
NOUN-AGRESS	dragon, monster
NOUN-FRAG	glass, plate
NOUN-FOOD	cookie, break
VERB-INTRAN	think, sleep
VERB-TRAN	see, chase
VERB-AGPAT	move, break
VERB-PERCEPT	smell, see
VERB-DESTROY	break, smash
VERB-EAT	eat

TABLE 4
Templates for Sentence Generator

WORD 1	WORD 2	WORD 3
NOUN-HUM	VERB-EAT	NOUN-FOOD
NOUN-HUM	VERB-PERCEPT	NOUN-INANIM
NOUN-HUM	VERB-DESTROY	NOUN-FRAG
NOUN-HUM	VERB-INTRAN	
NOUN-HUM	VERB-TRAN	NOUN-HUM
NOUN-HUM	VERB-AGPAT	NOUN-INANIM
NOUN-HUM	VERB-AGPAT	
NOUN-ANIM	VERB-EAT	NOUN-FOOD
NOUN-ANIM	VERB-TRAN	NOUN-ANIM
NOUN-ANIM	VERB-AGPAT	NOUN-INANIM
NOUN-ANIM	VERB-AGPAT	
NOUN-INANIM	VERB-AGPAT	
NOUN-AGRESS	VERB-DESTROY	NOUN-FRAG
NOUN-AGRESS	VERB-EAT	NOUN-HUM
NOUN-AGRESS	VERB-EAT	NOUN-ANIM
NOUN-AGRESS	VERB-EAT	NOUN-FOOD

Man eat cookie
Woman smell book
Mouse chase mouse
Dragon smash glass



Vector promedio de activaciones de capa oculta para cada palabra

Agrupados según similitud

Red recurrente de Elman

- Capaz de aprender patrones de recurrencia
- Genera representaciones intermedias según patrones de ocurrencia
  - Según “categoría gramatical”

Cómo se estudian los fenómenos **mentales**?

1. Comportamiento
2. Mirar el cerebro (neuroimágenes)

1. **Comportamiento**
2. Mirar el cerebro (neuroimágenes)

Comportamiento

- Verbal
  - Cuestionarios
  - Tareas de producción: asociación libre, naming, fluencia verbal

- No-verbal
  - 2AFC: Tarea de decisión léxica, decisión semántica, etc.
  - Libre: lectura
  - Juicios: similitud

Asociación libre

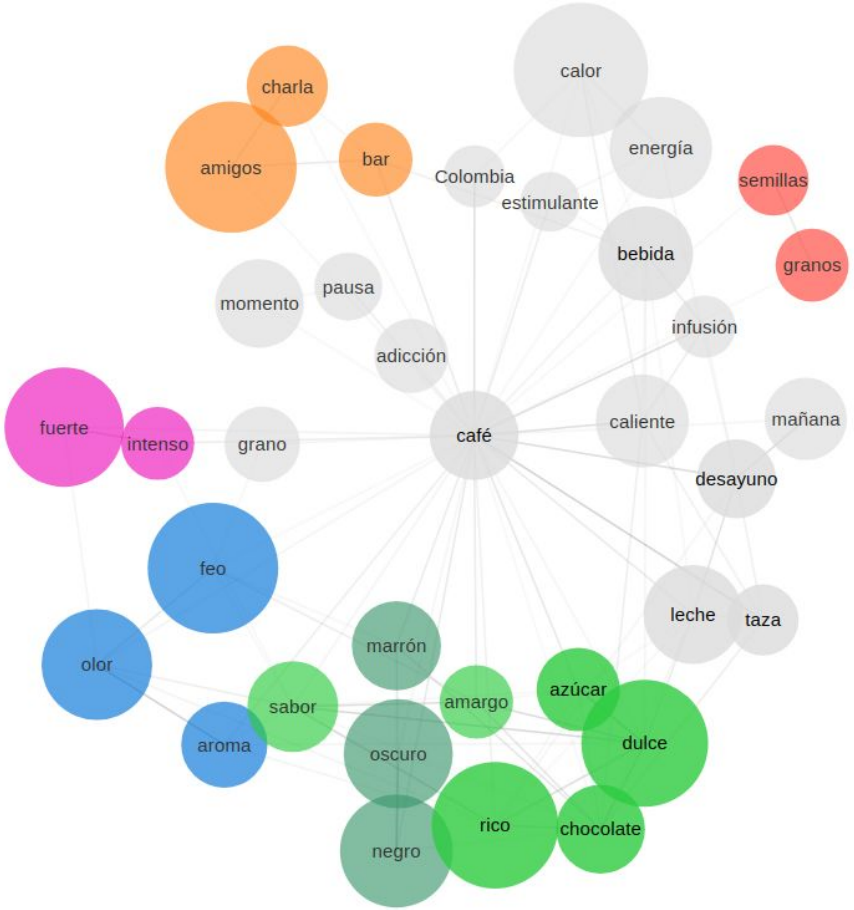
Si yo te digo “**token**”, vos pensás en **?**

clave (cue) → respuesta (response)

Normas de Asociac

Base de datos de clave

Grafo



Behavior Research Methods
https://doi.org/10.3758/s13428-023-02070-z

The "Small World of Words" free association norms for Rioplatense Spanish

Álvaro Cabana[1,3] · Camila Zugarramurdi[1,4] · Juan C. Valle-Lisboa[4,5] · Simon De Deyne[2]

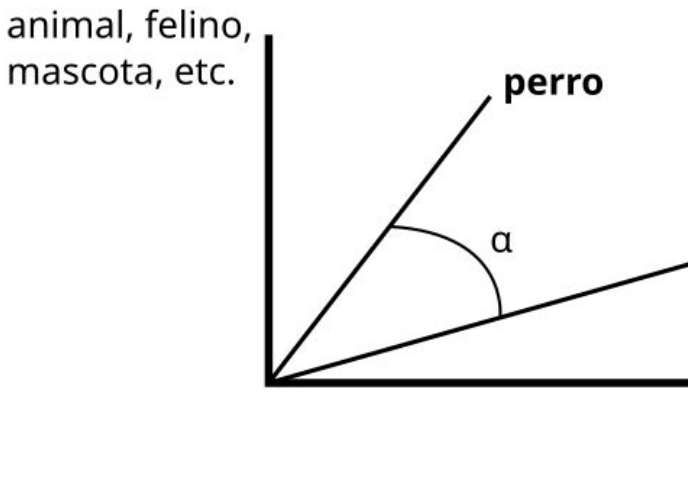
Accepted: 16 January 2023
© The Psychonomic Society, Inc. 2023

Una medida diferente de *frecuencia*

Tokens		Types	
R1	R123	R1	R123
agua [water]	agua [water]	amor [love]	amor [love]
comida [food]	comida [food]	trabajo [work]	trabajo [work]
amor [love]	amor [love]	persona [person]	vida [life]
animal [animal]	trabajo [work]	comida [food]	persona [person]
trabajo [work]	dolor [pain]	vida [life]	dolor [pain]
dolor [pain]	dinero [money]	dolor [pain]	comida [food]
dinero [money]	música [music]	yo [me]	familia [family]
música [music]	animal [animal]	malo [bad]	personas [persons]
malo [bad]	vida [life]	agua [water]	tiempo [time]
casa [house]	casa [house]	hombre [man]	malo [bad]

Word embeddings

- Espacios semánticos - representaciones espaciales del significado
- Cada palabra es un vector (punto en el espacio)
- Similitud en significado: cercanía en el espacio (o paralelismo)
- Típicamente calculados a partir de grandes colecciones de textos (**word2Vec**, **fastText**, **GloVE**)



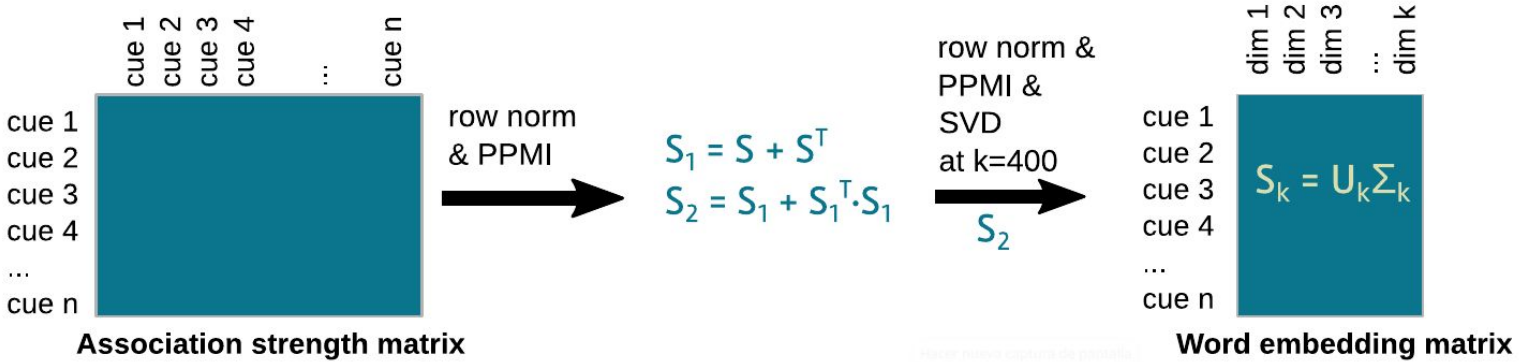
similarity(perro,gato) = cos(α)

Word Association Spaces

- **Word embeddings** a partir de normas de asociación

- Varios procedimientos

Word Association Spaces (WAS)



Cuestionarios: normas léxicas

- Pensá en la palabra “**chicharra**”... qué tan positiva o negativa es esa palabra?

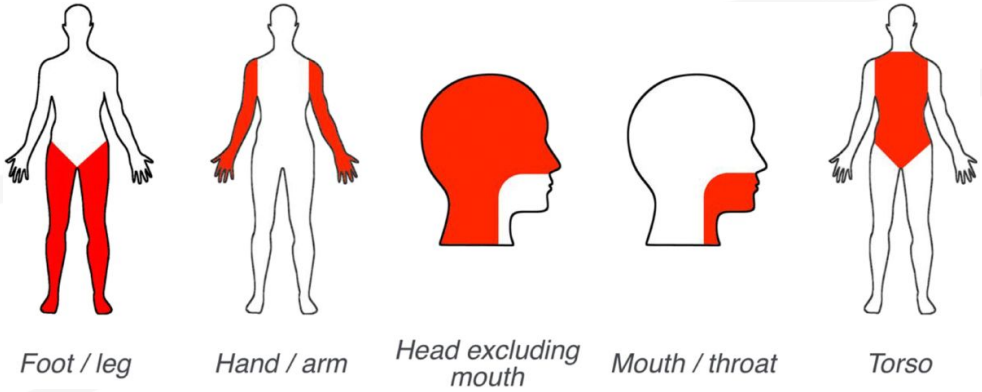
Word	Word_english	Val Mn
abandono	abandonment	1,47
abatir	to knock down	3,03
abierto	open	5,6
abismo	abyss	2,77
abrazar	to hug	8,67
abrazo	hug	8,14
abroncar	to get angry	2,43
abuso	abuse	1,47
acariciar	to caress	8,2
acatar	to comply with	3,97
aceptación	acceptance	7,77
acera	sidewalk	4,43
aclamar	to acclaim	6,13

1 muy

10 sitiva

Cuestionarios: normas léxicas

- **Valencia**: valor emocional positivo o negativo
- **Activación** (arousal): qué tan “activante” es una palabra (*almohada*, *vértigo*)
- **Iconicidad**: *bang*
- **Sensorio-motoras**:



Juicios de similitud

- Qué tan similares son (entre 0 y 1):
    - Gato - perro
    - Trompeta - trombón
    - Hueso - silla

- Elige al que es diferente
    - Arroz, fideos, garbanzos

Tarea de decisión léxica (LDT)

¿es una **palabra** o no?

fruta

análogo

brunalda

murciélago

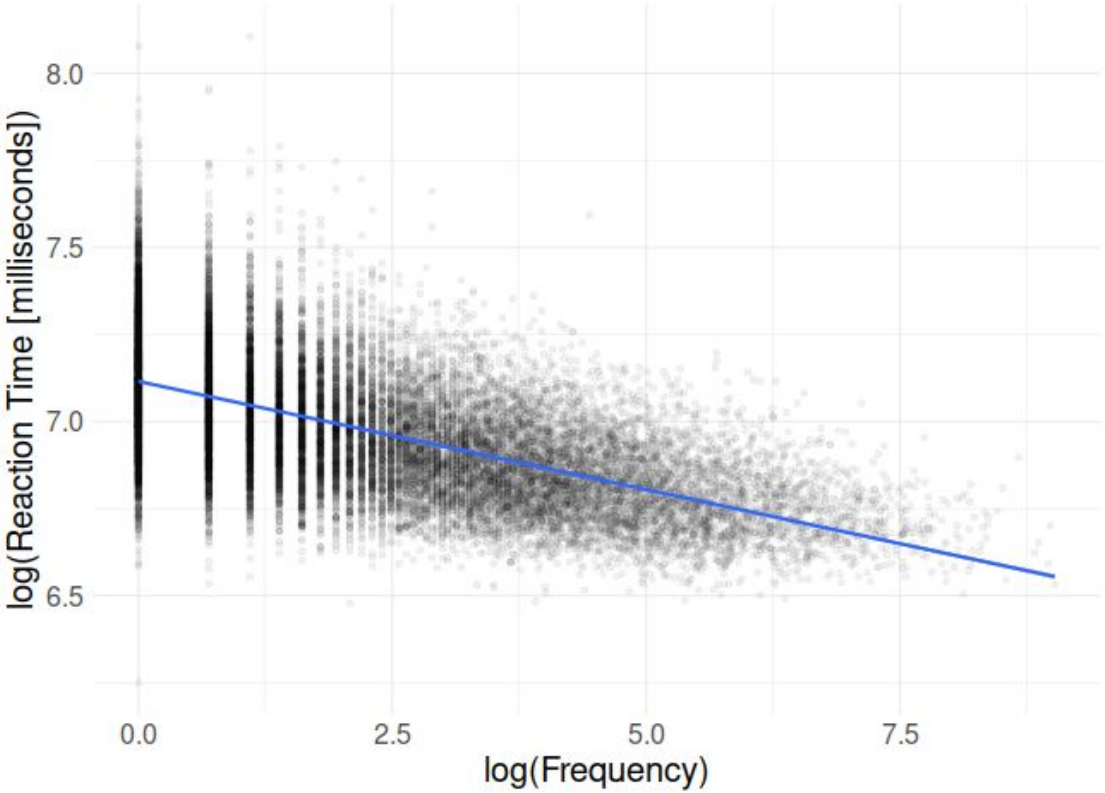
bruñir

escoba

ródula

muy bien!!

Tiempo de reacción en LDT y frecuencia



Frecuencia estimada del corpus SUBTLEX-ES. (Cuetos et al., 2009)

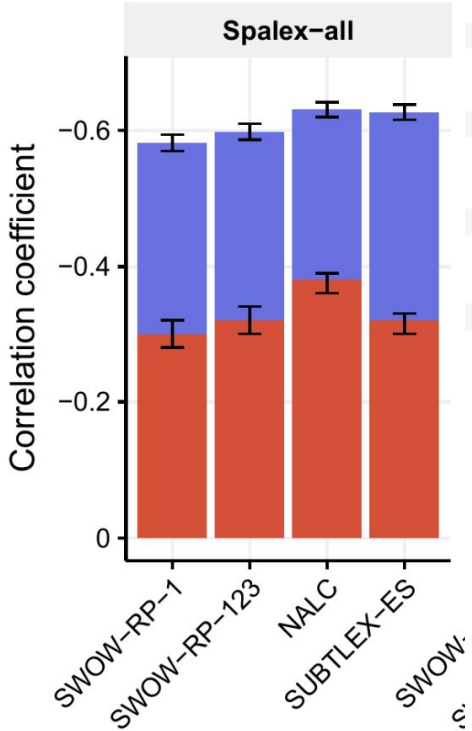
¿Qué pasa con las frecuencias de asociación?

¿Pueden obtenerse estimaciones de frecuencia léxica relevantes para el procesamiento a partir de asociación libre?

Tiempo de reacción en LDT y frecuencia

Respuesta corta: sí

Estimaciones de frecuencia basadas en asociación libre (SWOW, NALC) **explican** varianza única (incluso cuando se **parcializa** la contribución de normas de frecuencia basada en textos)

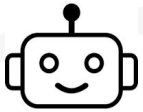


¿Cómo se comparan word embeddings?

- Basados en texto vs. basados en asociación libre

- Correlacionando con juicios de similitud semántica humana:
  - ¿qué tan similares son?





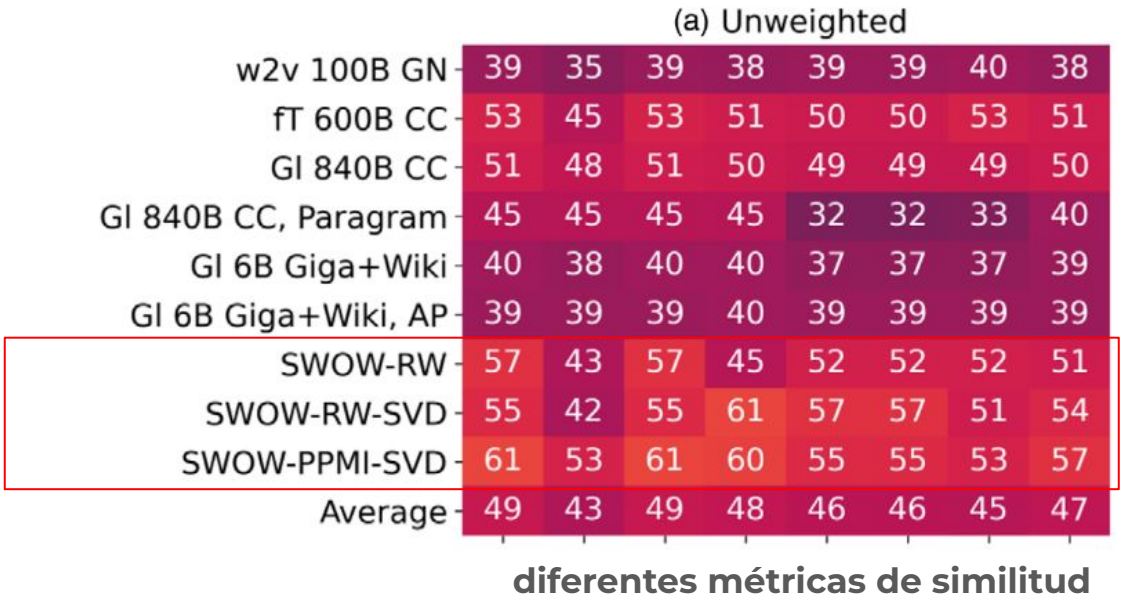
word 1	word 2		
gato	perro	0.8	0.7
gato	ratón	0.6	0.6
perro	cardenal	0.4	0.5
perro	ratón	0.6	0.5
ratón	cardenal	0.5	0.4



correlación

Comparando diferentes word embeddings con humanos

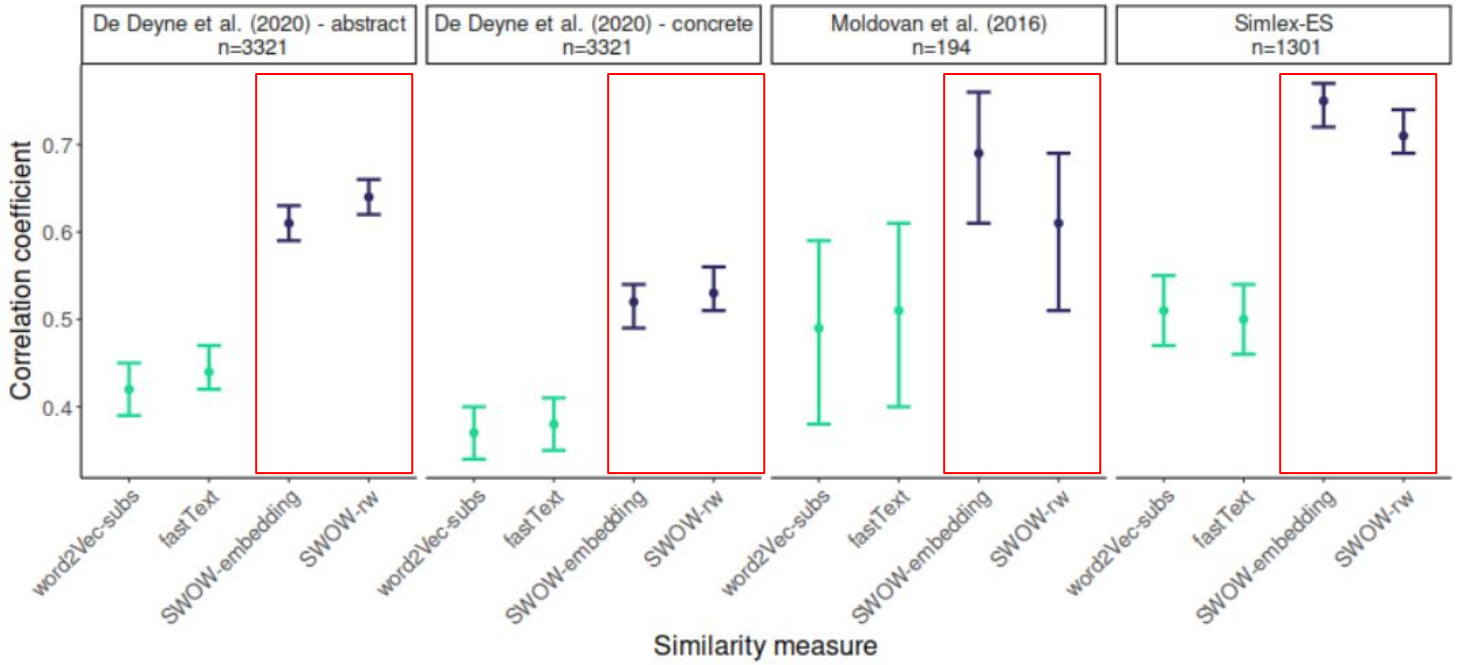
Mejor desempeño de embeddings basados en asociación libre



Richie & Bhatia, 2021

Comparando diferentes word embeddings con humanos

Español Rioplatense



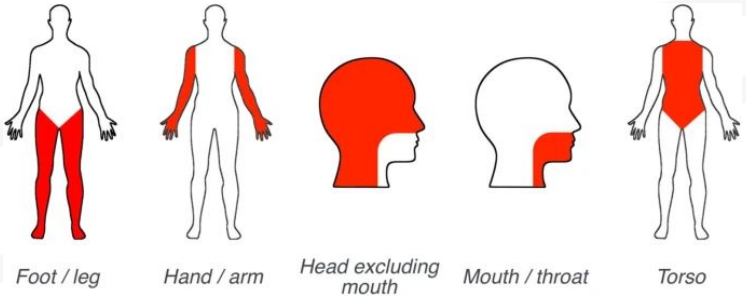
Cabana et al., 2024

Correlación con activación, valencia, normas somatosensoriales

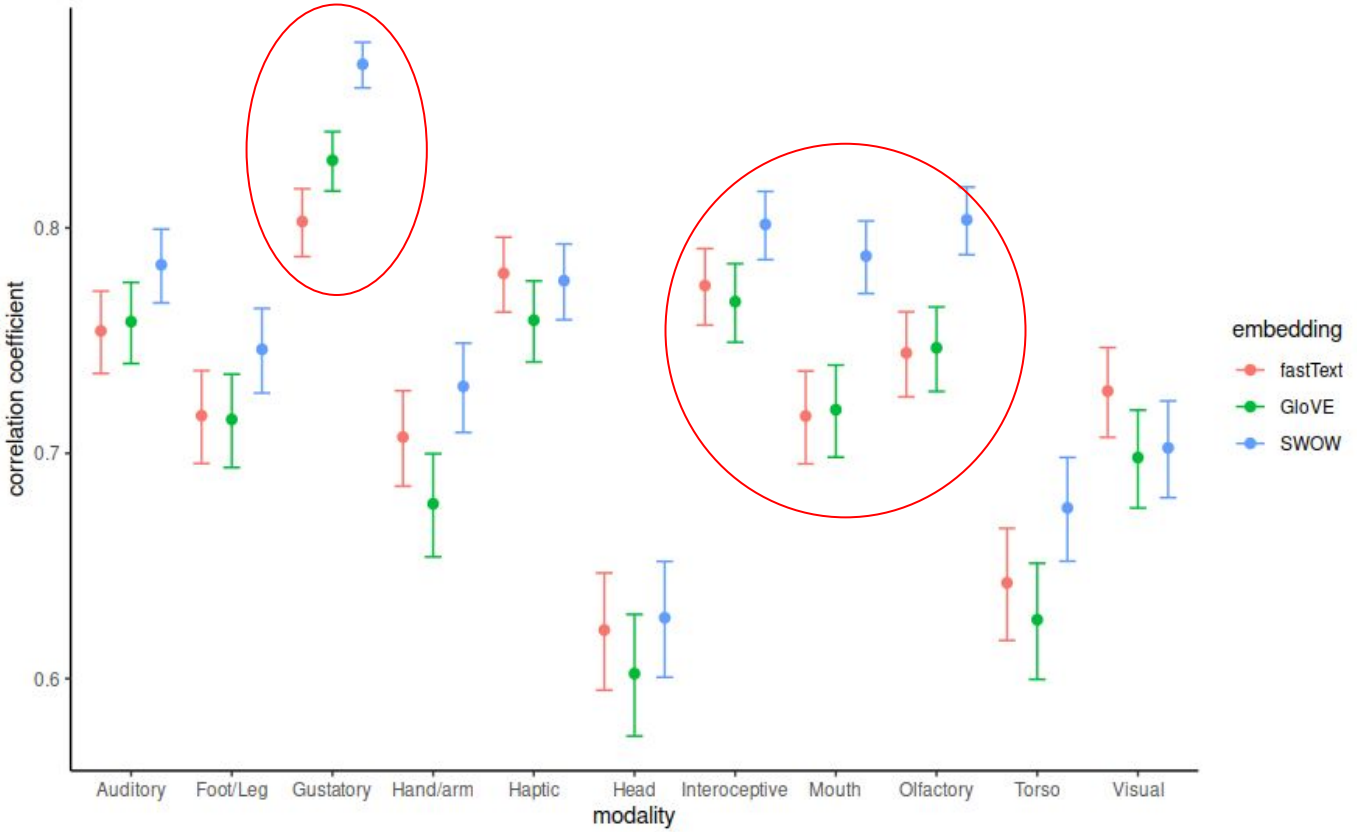
Home > Behavior Research Methods > Article

The Lancaster Sensorimotor Norms: multidimensional measures of perceptual and action strength for 40,000 English words

Open access | Published: 12 December 2019



Mejor predicción de normas somatosensoriales

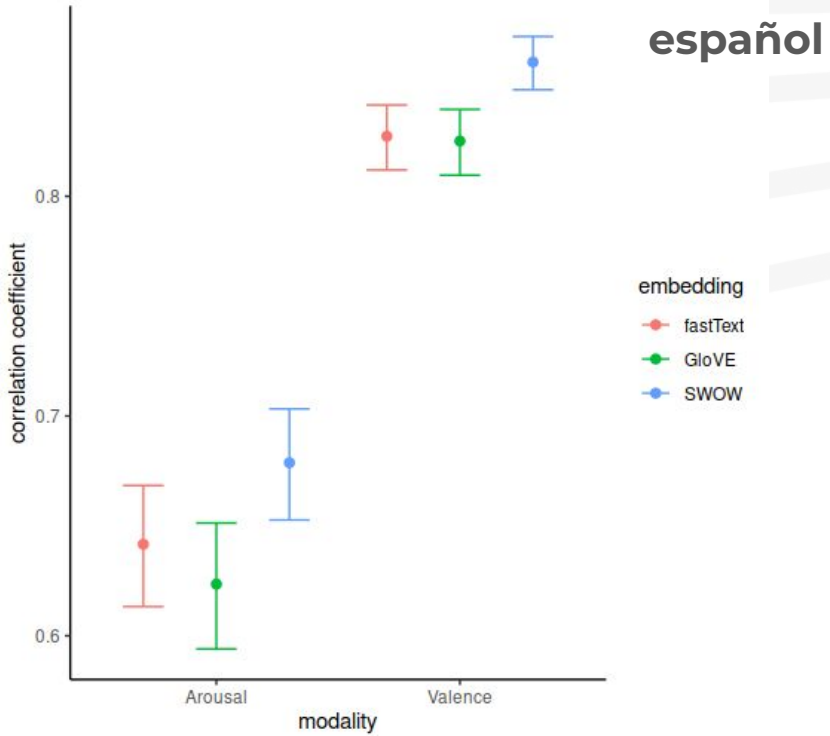


Mejor predicción de valencia y activación

En inglés, además:

Edad de adquisición, dominancia, concretitud

Vankrunkelsven et al., 2018



¿Y si le preguntamos a los LLMs?

Restricted access | Research article | First published online November 30, 2024 | Re-use permissions

AI-generated estimates of familiarity, concreteness, valence, and arousal for over 100,000 Spanish words

Gonzalo Martínez, Javier Conde, [...], and Marc Brysbaert +1 View all authors and affiliations

Volume 78, Issue 10 | https://doi.org/10.1177/17470218241306694 | View article versions

"¿Podrías calificar la concreción de la siguiente en una escala del 1 al 7, donde 1 significa muy abstracto y 7 significa muy concreto? Ejemplos de palabras que recibirían una calificación de 1 son espiritualidad, convencionalismo y creencia. Ejemplos de palabras que recibirían una calificación de 7 son trineo, margarita y pavo real. La expresión es: [insertar expresión aquí]. Solo responde con un número del 1 al 7. Por favor, limita tu respuesta a números."

Normas producidas por LLMs

- Alta correlación con juicios humanos (entre .7 y .9)
- Igual o mejor desempeño en tareas de validación (LDT, lectura, etc)
- Permiten extender normas a lenguas poco estudiadas
- Y a unidades supraléxicas…

Similitud ...

Large language models predict human sensory judgments across six modalities

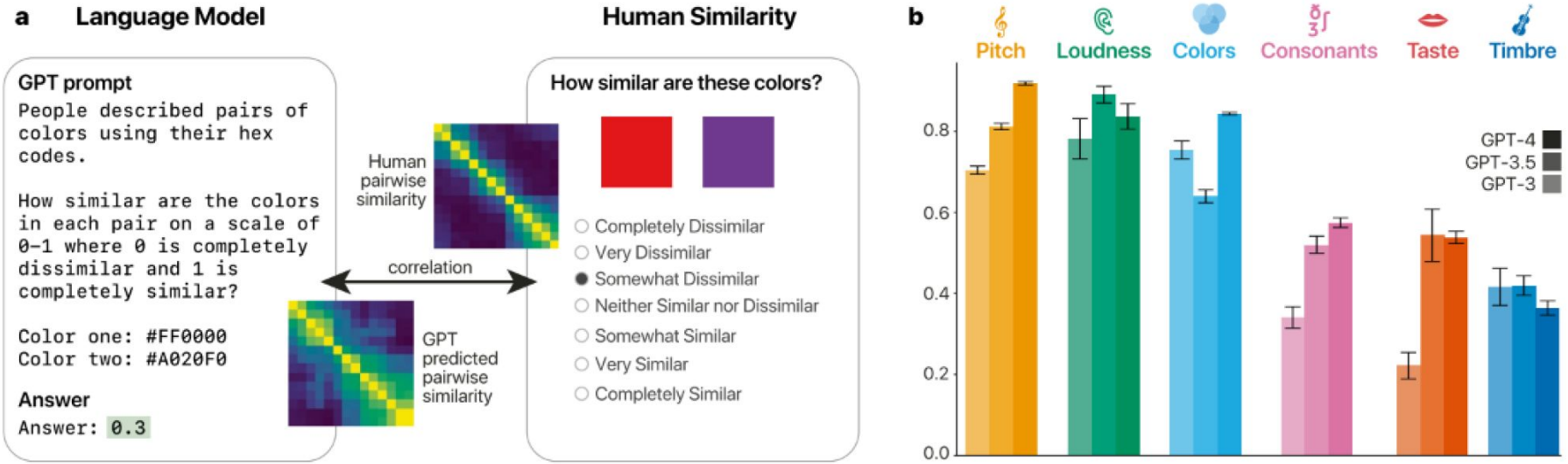
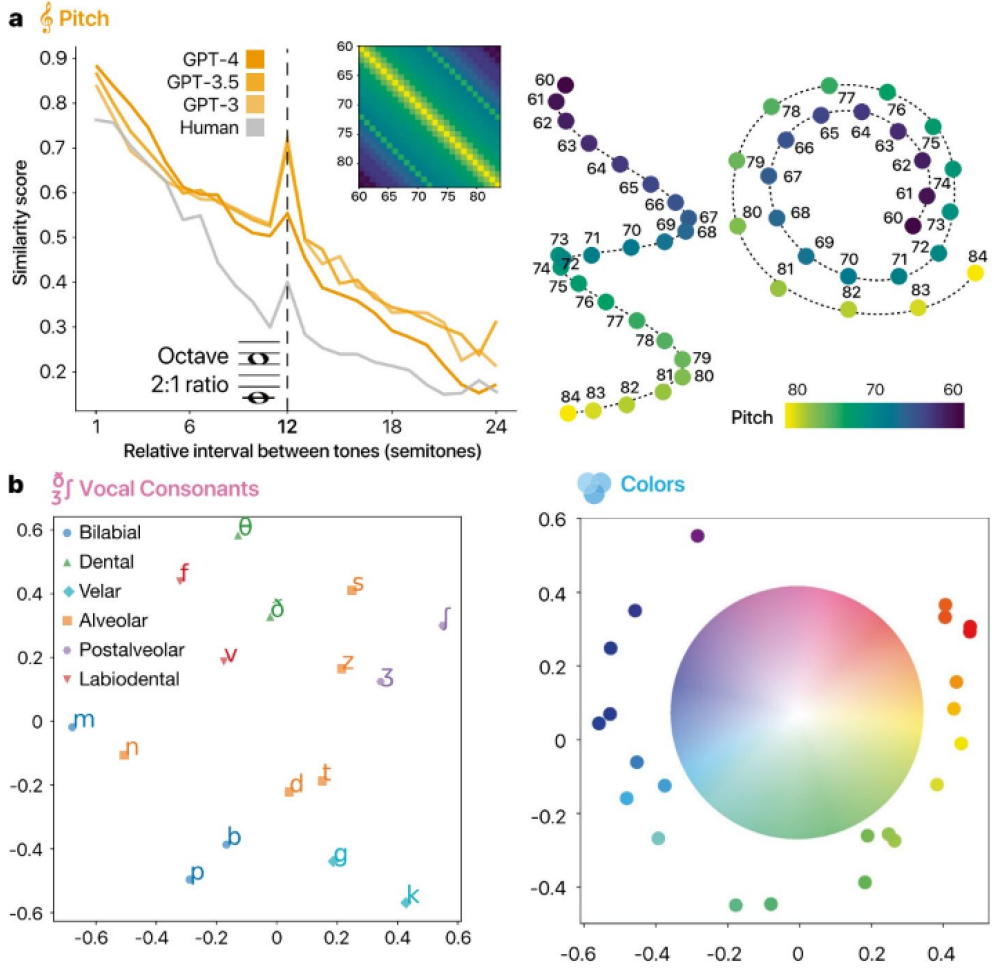


Fig. 1. (**a**) Schematic of the LLM-based and human similarity judgment elicitation paradigms. (**b**) Correlations between models and human data across six perceptual modalities, namely, pitch, loudness, colors, consonants, taste, and timbre (Pearson *r*; 95% CIs).

MDS


a
Pitch
GPT-4
GPT-3.5
GPT-3
Human
Similarity score
Octave
2:1 ratio
Relative interval between tones (semitones)
Pitch
b
Vocal Consonants
Bilabial
Dental
Velar
Alveolar
Postalveolar
Labiodental
Colors

LLMs y asociación libre...

The "LLM World of Words" English free association norms generated by large language models

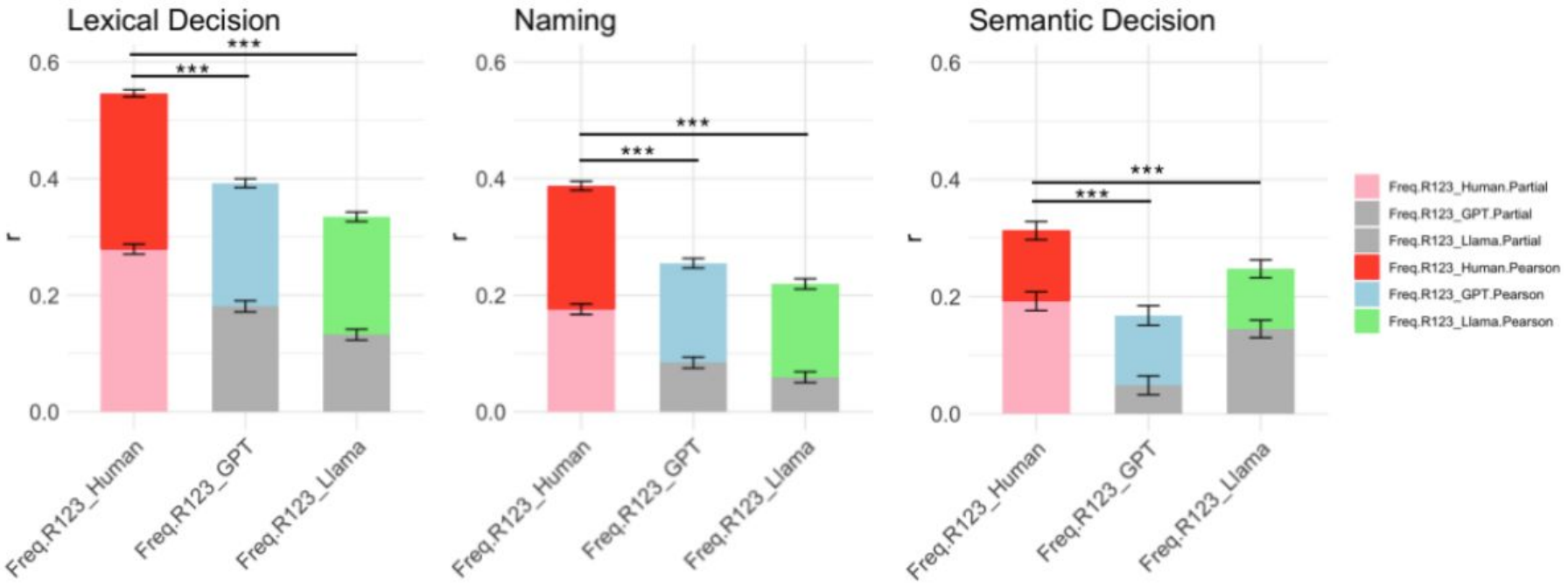
Katherine Abramski ✉, Riccardo Improta, Giulio Rossetti & Massimo Stella ✉

Scientific Data **12**, Article number: 803 (2025) | Cite this article

Human-likeness of LLMs in the Mental Lexicon

Bei Xiao[1] Xufeng Duan[1] David A. Haslett[2] Zhenguang G. Cai[1,3]
[1]Department of Linguistics and Modern Languages, The Chinese University of Hong Kong
[2]Division of Social Science, The Hong Kong University of Science and Technology
[3]Brain and Mind Institute, The Chinese University of Hong Kong
BeiXiao@link.cuhk.edu.hk

La red de asociación de los LLMs....



Usar asociación libre para hacer *fine tuning?*

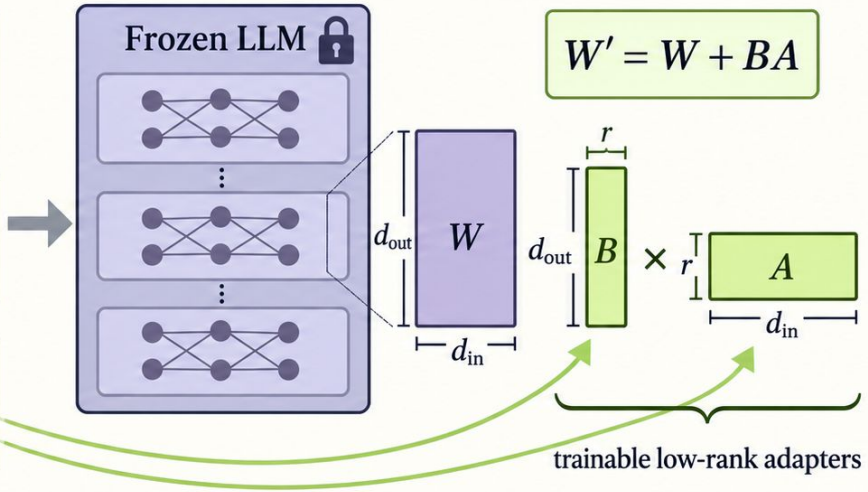
SWOW training examples

Cue: dog
Associates: cat, pet, bark, puppy, leash

Cue: apple
Associates: fruit, red, tree, pie, food

Cue: chair
Associates: table, sit, furniture, seat, desk

LoRA fine-tuning



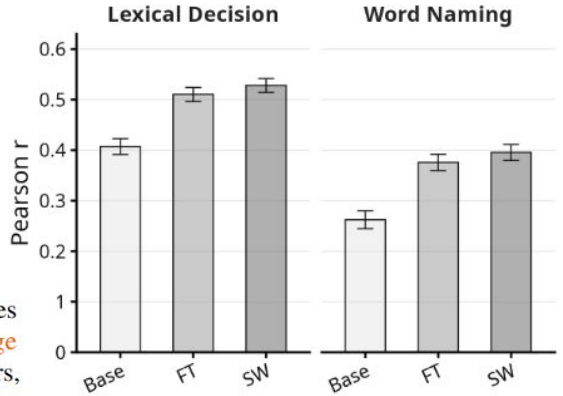
Rank	food			raven			anti			justice		
	Base	FT	SWOW	Base	FT	SWOW	Base	FT	SWOW	Base	FT	SWOW
1	pizza	eat	eat	intelligence	bird	black	vaccine	against	against	fairness	judge	court
2	salad	meal	good	mystery	black	bird	virus	negative	establishment	equality	court	law
3	restaurant	hungry	sustenance	midnight	crow	crow	antibody	opposite	opposite	morality	fair	fair
4	apple	hunger	meal	intelligent	poe	poe	gravity	war	con	law	law	judge
5	sandwich	drink	cook	death	murder	poem	bacterial	social	no	laws	right	peace

Table 2: Agreement between the Llama-base (Base) and Llama-FT (FT) and SWOW (SW) associations across lexico-semantic, distributional and graph-based measures. $\bar{X}_{SW}$ and $\bar{X}_{LLM}$ denote mean values. Results show the means, Pearson correlation r and ρ_c Lin's concordance coefficient (width of 95% CI for r and ρ_c all below .04, highest value in bold).

		Base			FT		
Measure	$\bar{X}_{SW}$	$\bar{X}_{LLM}$	r	ρ_c	$\bar{X}_{LLM}$	r	ρ_c
Word length	6.19	7.22	.71	.43	6.03	.83	**.80**
Noun prob	0.67	0.57	.80	.70			
log(WF)	3.03	2.71	.72	.51	3		
AoA	6.62	7.38	.82	.66	6		
Valence	5.30	5.45	.92	.87	5		
Arousal	3.30	3.46	.91	.88	3		
Concreteness	2.86	2.92	.91	.88	2		

Table 3: Pearson correlations (r) between model similarities and human similarity ratings. Within each measure, orange indicates a correlation significantly larger than both others, and purple indicates a correlation significantly smaller than both others (pairwise dependent Steiger's Z tests, $p < .05$).

		PPMI			RW		
Dataset	n	Base	FT	SW	Base	FT	SW
MEN	1,793	.664	.725	.691	.818	.834	.804
MTURK	478	.610	.696	.668	.730	.771	.764
Silberer14	4,085	.750	.786	.761	.818	.848	.815
SimLex	630	.592	.637	.634	.654	.662	.694
SimVerb	2,227	.488	.558	.557	.540	.607	.637
WS Cloze	222	.607	.641	.636	.668	.691	.735
WS Taxonomic	222	.708	.760	.721	.791	.826	.843
WS Thematic	222	.675	.737	.683	.779	.818	.823

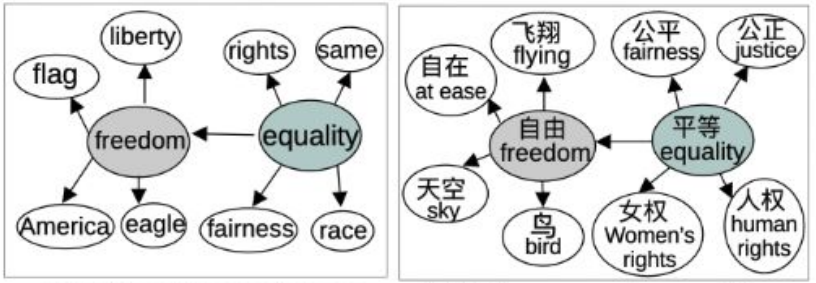


Correlations with lexical decision and word naming
RTs for the Base, FT and SWOW (SW) models.
ests for dependent overlapping correlations showed
e correlations were significantly different.

ALIGN: Word Association Learning for Cultural Alignment in Large Language Models

Chunhua Liu[*1] **Kabir Manandhar Shrestha**[*2] **Sukai Huang**[*3]
[1]School of Computing and Information Systems, The University of Melbourne
[2]Melbourne Data Analytics Platform, The University of Melbourne
[3]Faculty of Information Technology, Monash University
chunhua.liu1@unimelb.edu.au
k.manandharshrestha@unimelb.edu.au
sukai.huang@monash.edu

(1) Lexical-level Cultural Associations



(a) US word associations (b) Chinese word associations

(2) High-level Cultural Value

Test	Metric	Human	$\text{Llama}_{\text{van}}$	$\text{Llama}_{\text{ppo}}$	$\text{Llama}_{\text{sft}}$	Qwen_{van}	Qwen_{ppo}	Qwen_{sft}
SWOW.US	Valence	5.514	5.398	5.403	**5.543***	5.337	5.352	**5.484***
	Arousal	4.244	**4.272***	**4.238***	4.214	4.192	4.183	4.192
	Concreteness	3.644	3.378	3.355	3.582	3.368	3.349	3.535
	Emotional %	84.6%	78.2%	77.5%	75.5%	73.5%	73.5%	74.9%
	%Conc | %Abs | %Unk	64.3/29.8/5.9	52.8/37.9/9.3	51.1/38.7/10.2	56.8/29.0/14.2	50.5/37.0/12.5	50.4/37.2/12.4	56.7/29.6/13.7
SWOW.ZH	Valence	5.386	5.341	5.311	**5.427***	5.352	5.332	**5.411***
	Arousal	5.378	5.258	5.270	**5.408***	5.233	5.220	**5.370***
	Concreteness	3.657	3.370	3.394	3.576	3.391	3.412	3.516
	Emotional %	53.3%	31.8%	33.8%	41.9%	42.3%	41.6%	47.9%
	%Conc | %Abs | %Unk	35.9/15.8/48.3	17.9/12.7/69.4	19.3/13.2/67.5	27.6/13.2/59.2	24.1/16.6/59.3	24.2/15.8/60.0	30.4/15.9/53.8

Table 3: Emotion and concreteness scores on SWOW.US (top) and SWOW.ZH (bottom). * Bold indicates no significant difference from human medians ($p \geq 0.05$).

Test	M Type	M Class	$\text{Train}_{\text{swow}}$	JSD↓	EMD↓
WVS.US	Vanilla	Llama	-	0.324	0.102
	SFT	Llama	US	0.392	0.114
	PPO	Llama	US	**0.288***	0.092
	Vanilla	Qwen	-	0.388	0.131
	SFT	Qwen	US	0.355*	0.118
	PPO	Qwen	US	0.353*	0.125
	Vanilla	Llama3.1_70b	-	0.294*	0.094
	Vanilla	Qwen2.5_72b	-	0.262*	0.109*
WVS.CN	Vanilla	Llama	-	0.459	0.152
	SFT	Llama	ZH	0.421*	0.129*
	PPO	Llama	ZH	0.334*	0.112*
	Vanilla	Qwen	-	0.415	0.139
	SFT	Qwen	ZH	**0.325***	**0.100***
	PPO	Qwen	ZH	0.374*	0.123*
	Vanilla	Llama3.1_70b	-	0.333*	0.100*
	Vanilla	Qwen2.5_72b	-	0.328*	0.116*

Table 4: World Values Survey results on US and CN. ↓ indicates that lower is better (higher alignment). * indicates the improvement over Vanilla is significant ($p < 0.05$).Prompting language matches the survey language (EN for US, ZH for CN).

Comportamiento y LLMs…

- Los LLMs son excelentes modelos del comportamiento verbal humano…

- Esta es sólo una muy sesgada muestra de cómo se pueden usar LLMs en psicolingüística
- Hay toda una literatura sobre la evaluación cognitiva de LLMs (p. ej.; teoría de la mente)
- Toda otra literatura sobre LLMs como modelos cognitivos más allá de aspectos léxicos…

1. Comportamiento
2. **Mirar el cerebro (neuroimágenes)**

Técnicas imagenológicas

EEG - electroencefalografía

ECOG - electrocórticoencefalografía

MEG - magnetoencefalografía

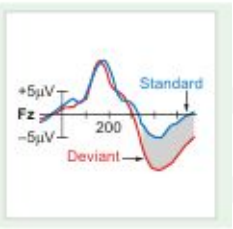
fMRI - resonancia magnética nuclear funcional

PET – tomografía emisión de positrones

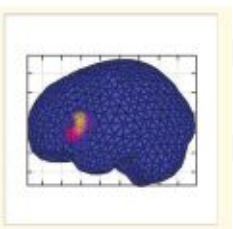
NIRS – espectroscopía del infrarrojo cercano

SPECT - tomografía de fotón único

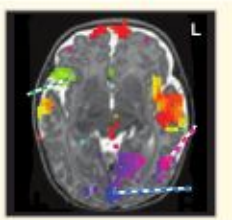


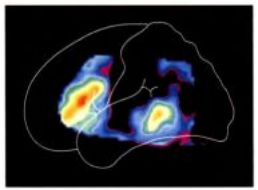


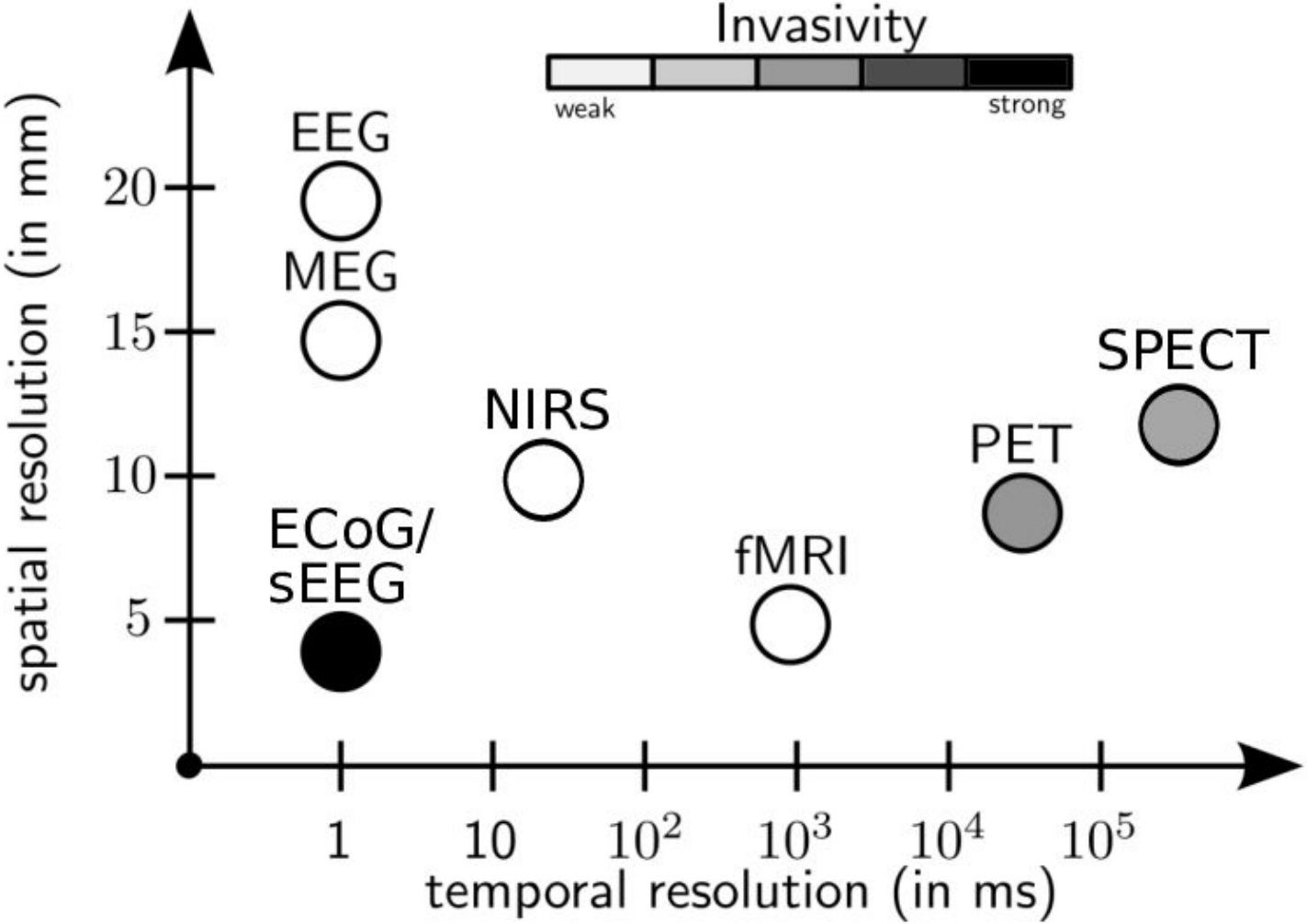




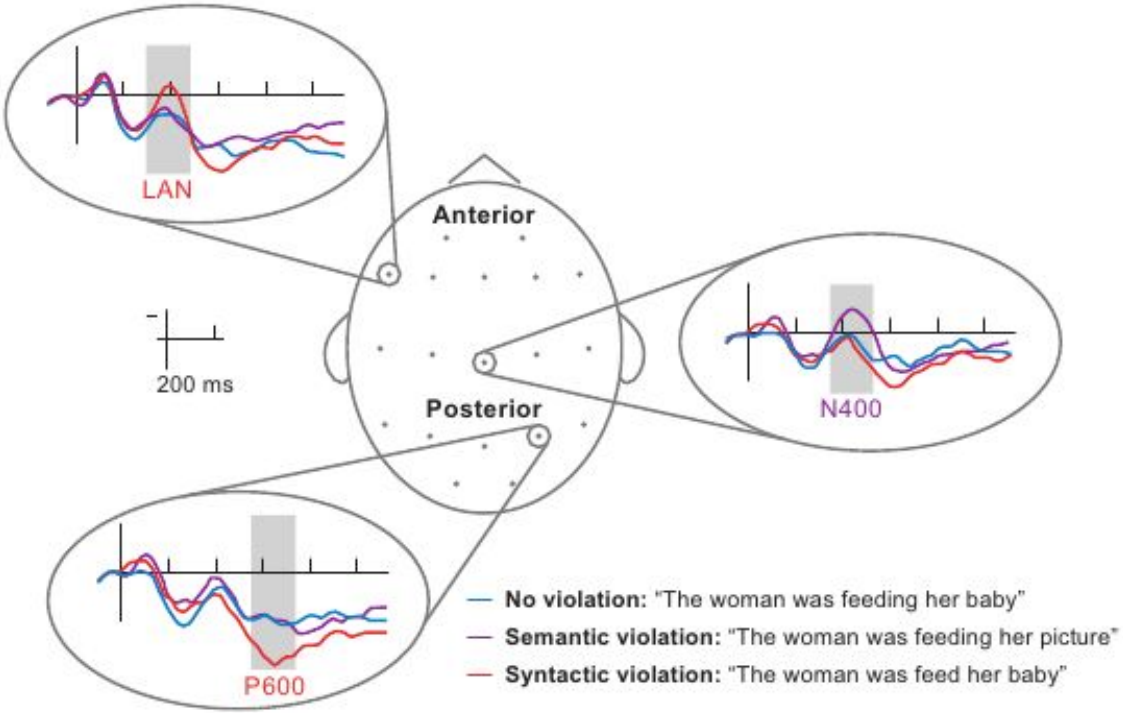







Invasivity
weak
strong
EEG
MEG
NIRS
ECoG/
sEEG
fMRI
PET
SPECT
spatial resolution (in mm)
20
15
10
5
1
10
10^2
10^3
10^4
10^5
temporal resolution (in ms)

EEG: Potenciales relacionados a eventos



Semántica – EEG

- Componente N400
  - Mayor amplitud
    - Palabras que “no encajan”
  - Integración en contexto semántico



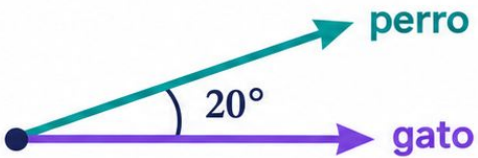
  - El futbolista pateó fuerte la *pelota*
  - El futbolista pateó fuerte la *almohada*

Cómo se pueden **usar** los modelos de lenguaje par estudiar los fenómenos **mentales**?

Métricas y predictores en LLMs (y word embeddings)

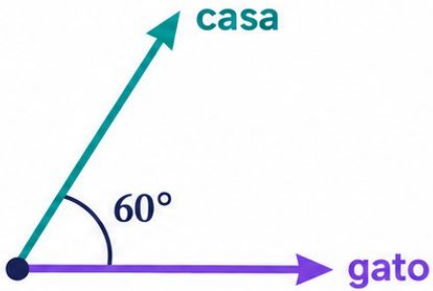
- Similitud (coseno, distancia euclídea, etc.)
- Sorpresa en la predicción: surprisal, entropía
- Embeddings de capas intermedias

Similitud entre palabras



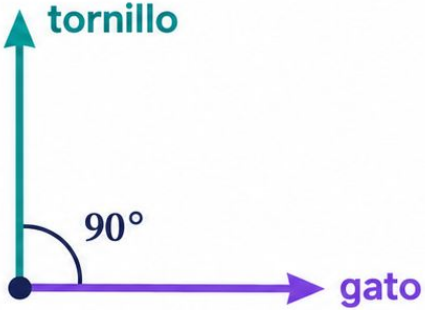
$\cos\theta \approx 0{,}94$

Direcciones parecidas



$\cos\theta = 0{,}50$

Menor similitud



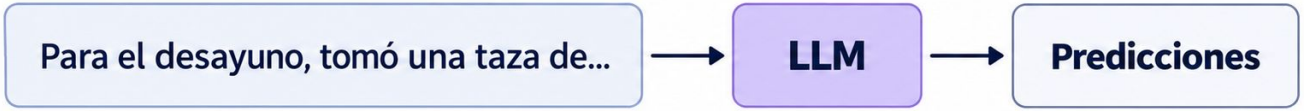
$\cos\theta = 0$

Direcciones ortogonales

$$\cos\theta = \frac{u \cdot v}{\|u\|\,\|v\|}$$

Ejemplo esquemático: los ángulos no provienen de un modelo real

Surprisal: ¿qué tan inesperada es una palabra?

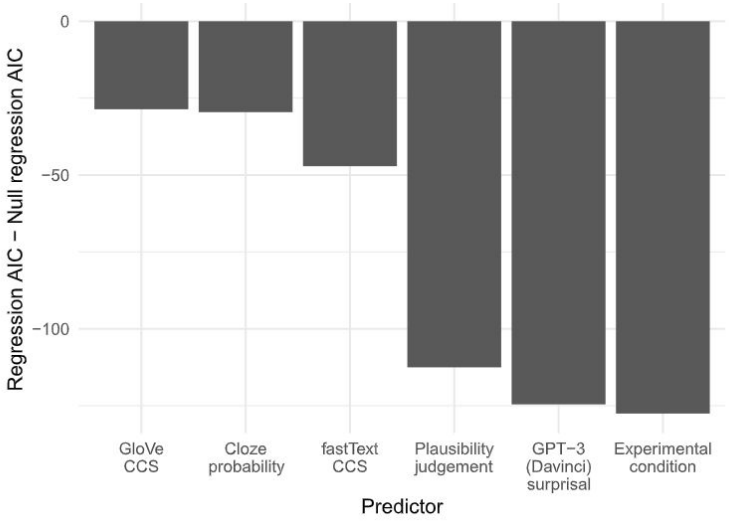


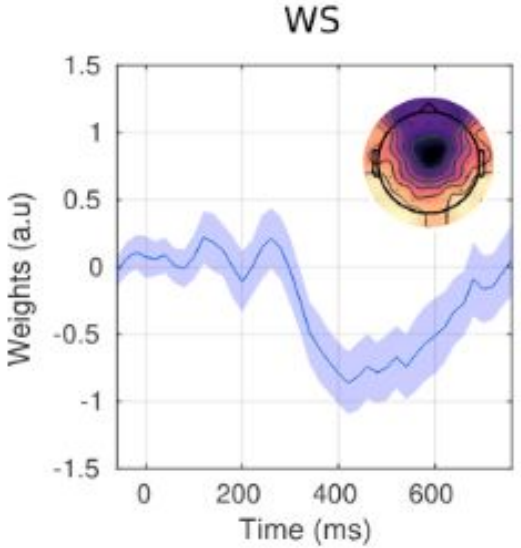
café	té	arena
p = 0,50	p = 0,25	p = 0,001
1 bit	2 bits	≈ 10 bits
Más esperable	Menos esperable	Muy inesperada

$$\text{surprisal}(w) = -\log_2 p(w \mid \text{contexto})$$

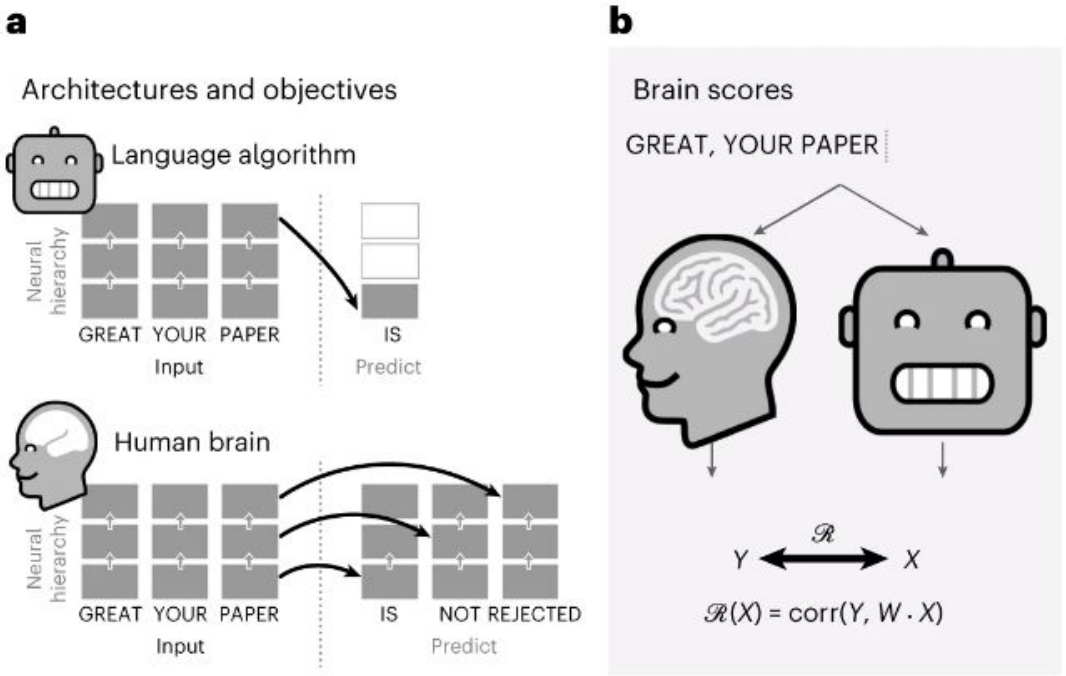
Probabilidades ilustrativas · ejemplo con palabras de un solo token

Word surprisal y el N400

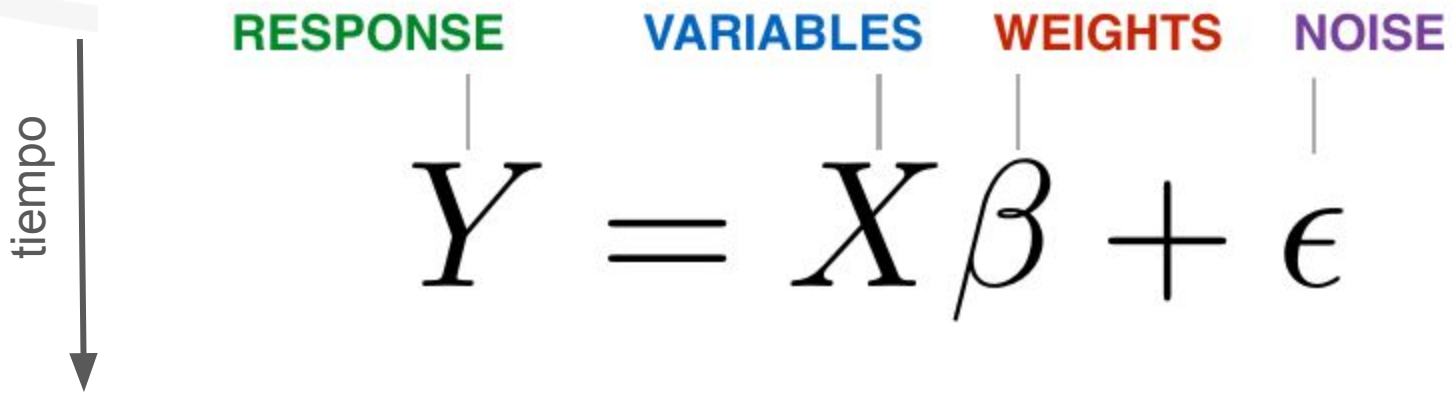




Se puede predecir la actividad cerebral con un LLM?

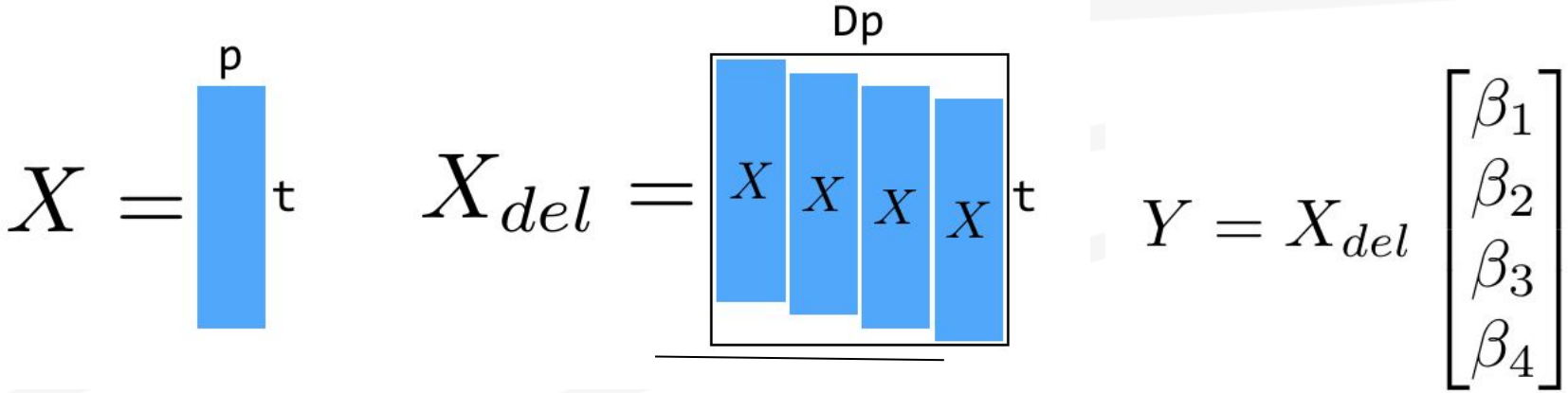


Para cada vóxel: regresión lineal



Filtro de Impulso Finito (FIR)

Concatenar copias retrasadas en el tiempo de la matriz de estímulos



$$X = \quad X_{del} = \quad Y = X_{del} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{bmatrix}$$

$$Y(t) = \beta_1 X(t) + \beta_2 X(t-1) + \beta_3 X(t-2) + \beta_4 X(t-3)$$

Regresión regularizada

RESPONSE VARIABLES WEIGHTS NOISE

$$Y = X\beta + \epsilon$$

$$Y_{t,j} \sim \mathcal{N}(X\beta, \sigma^2)$$

$$\hat{\beta} = \underset{\beta}{\operatorname{argmax}}\ P(Y|X,\beta)$$

Regresión regularizada

Introducir distribución *a priori*

RESPONSE VARIABLES WEIGHTS NOISE

$$Y = X\beta + \epsilon$$

$$Y_{t,j} \sim \mathcal{N}(X\beta, \sigma^2)$$

$$\hat{\beta} = \underset{\beta}{\operatorname{argmax}}\ P(Y|X,\beta)P(\beta)$$

Regresión regularizada

Penalización: restringir la norma de beta

RESPONSE VARIABLES WEIGHTS NOISE

$$Y = X\beta + \epsilon$$

$$E_{pen}(\beta) = ||Y - X\beta||_2^2 + \lambda||\beta||_2^2$$

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}}\ E_{pen}(\beta)$$

Regresión ridge

Penalización: restringir la norma de beta. O introducir una *a priori* Normal(0,σ^2I)

$$Y = X\beta + \epsilon$$

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \left[||Y - X\beta||_2^2 + \lambda ||\beta||_2^2 \right]$$

ERROR or LOSS PENALTY

“Now this is a story all about how my life got flipped-turned upside down…”



words

syntax

semantics

narrative

emotion

waveform

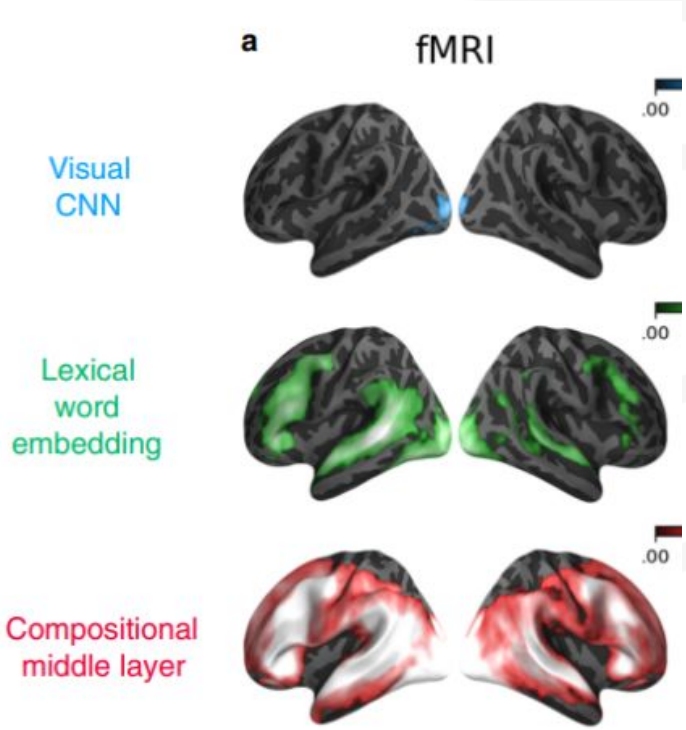
intonation

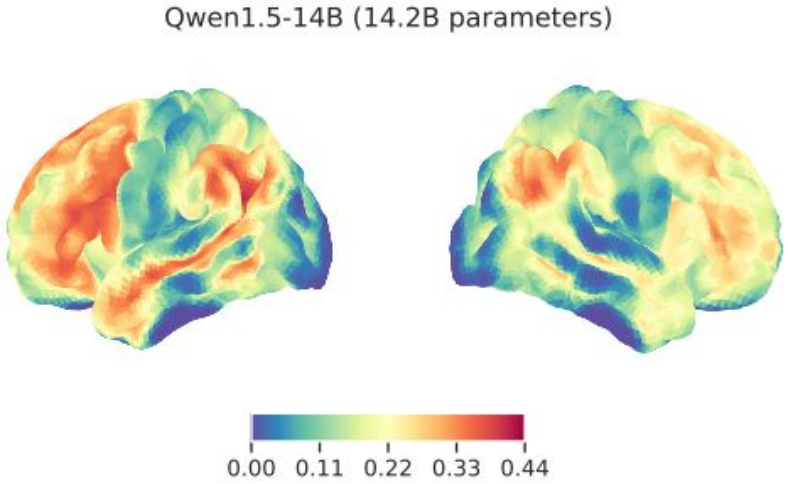
spectrum

phonemes

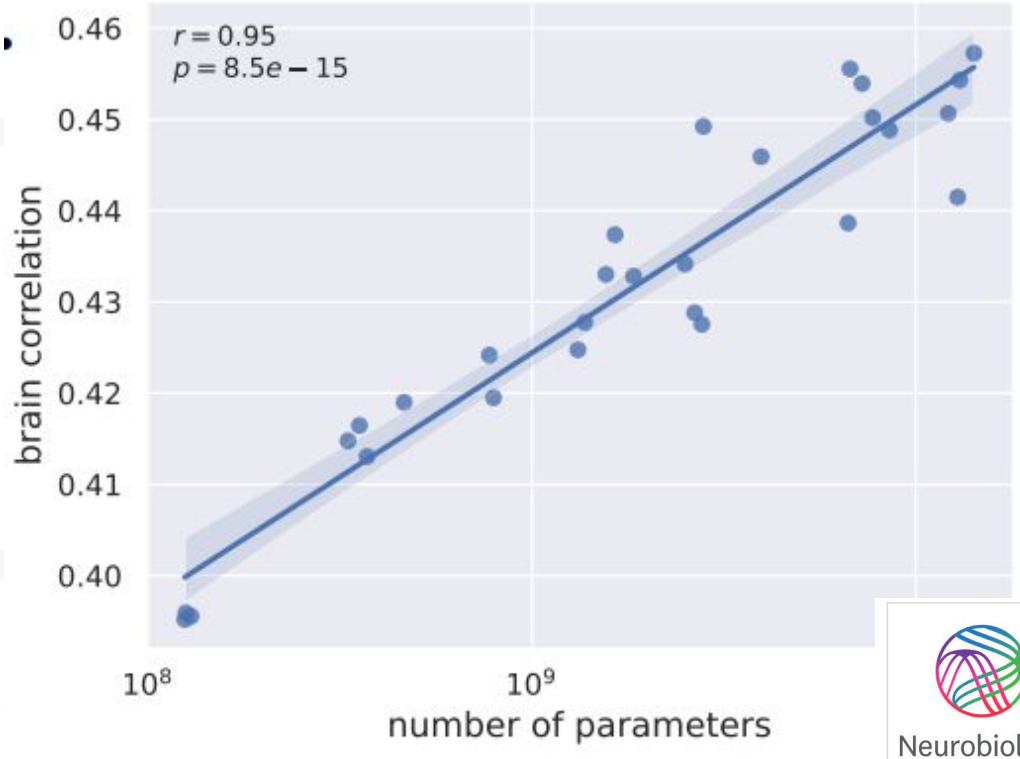
articulation

Los LLMs predicen mejor la activación





Billetera mata galán: más parámetros, mejor desempeño



fMRI predictors based on language models of increasing complexity recover brain left lateralization

Laurent Bonnasse-Gahot
Centre d'Analyse et de Mathématique Sociales
CNRS, EHESS
75006 Paris, France
lbg@ehess.fr

Christophe Pallier
Cognitive Neuroimaging Unit
CNRS, INSERM, CEA, Neurospin center
91191 Gif-sur-Yvette, France
christophe@pallier.org

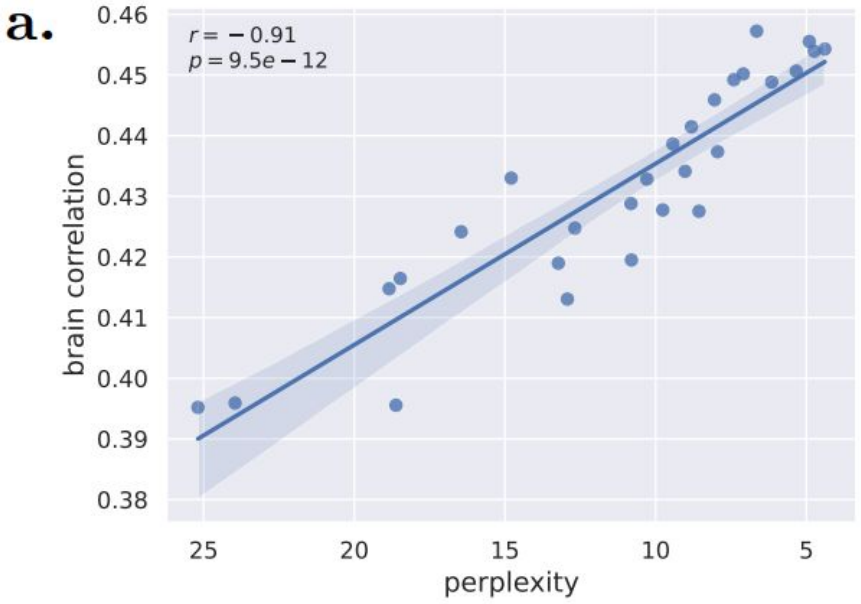
RESEARCH ARTICLE

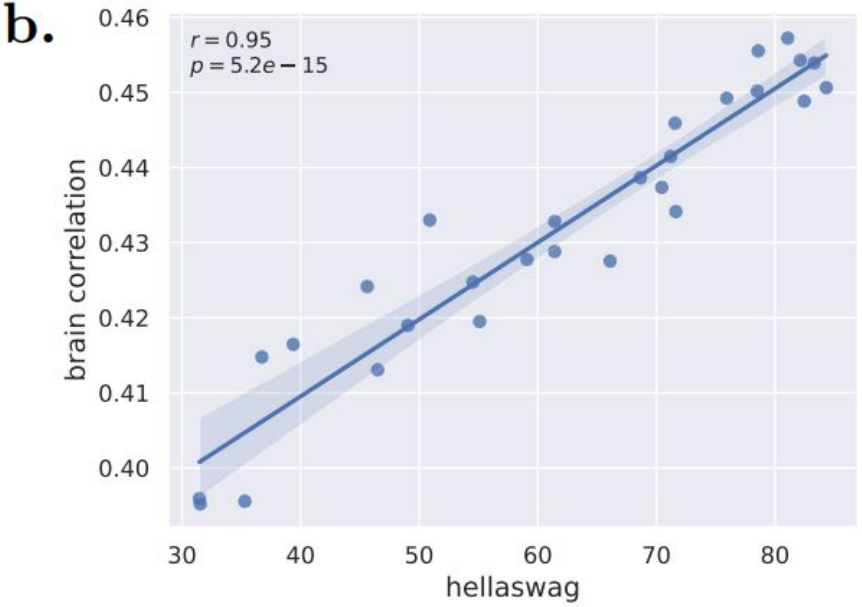
Left-Right Asymmetry in Predicting Brain Activity From LLMs' Representations Emerges With Their Formal Linguistic Competence

Laurent Bonnasse-Gahot[1] and Christophe Pallier[2]

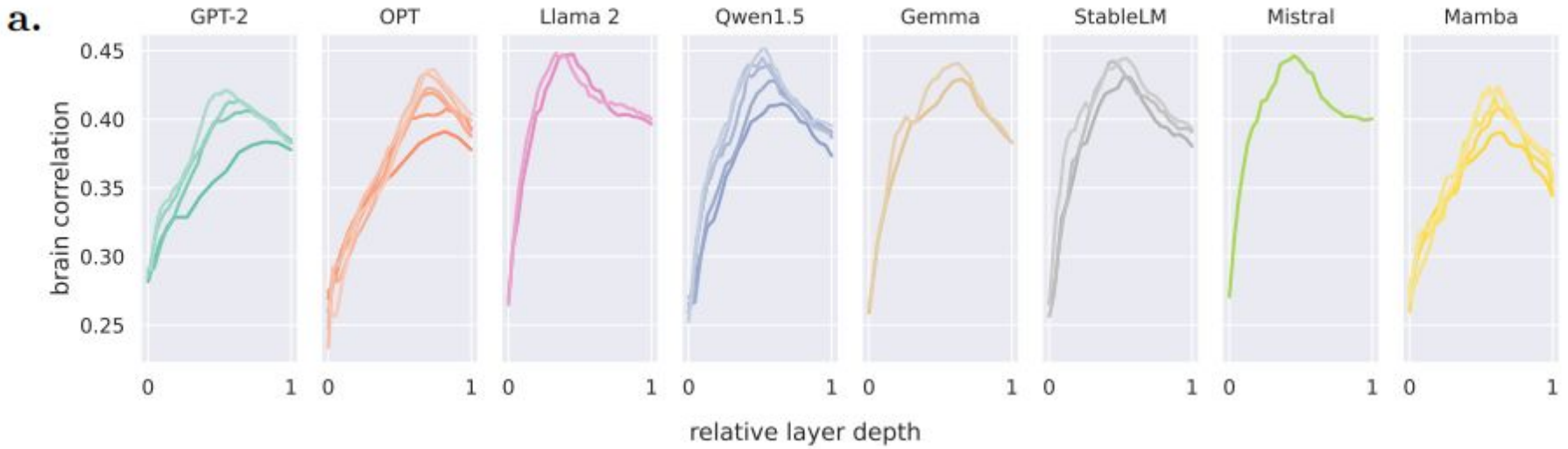
Neurobiology of Language

Mejores modelos, mayores correlaciones con actividad neural

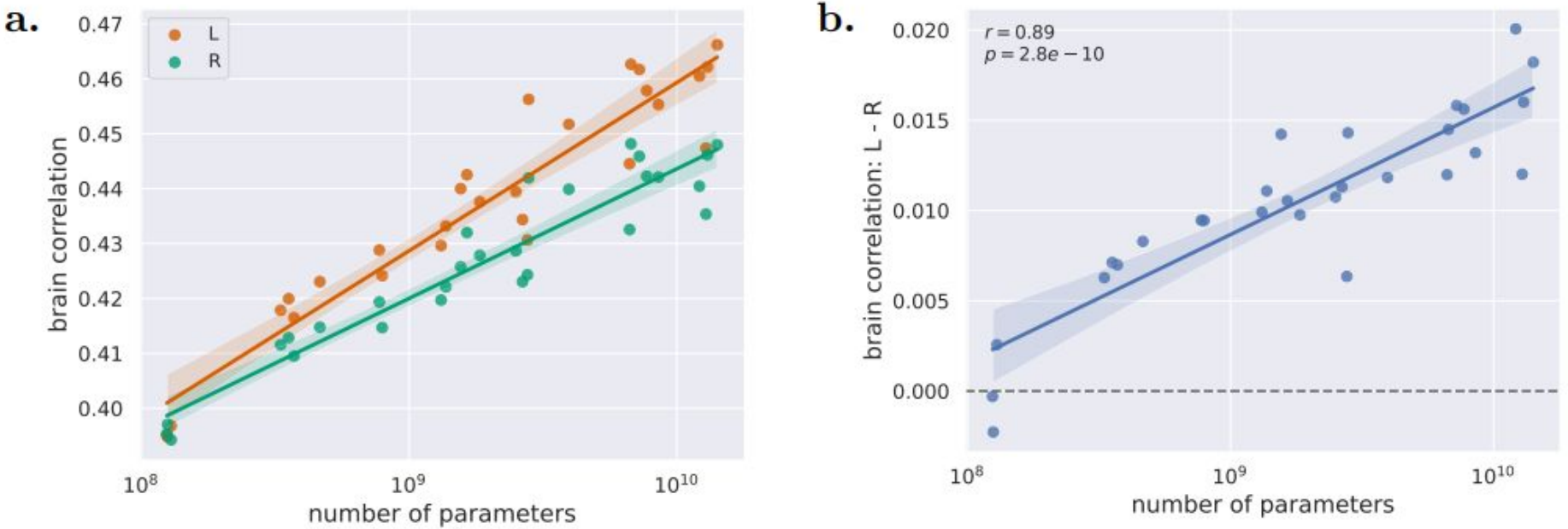




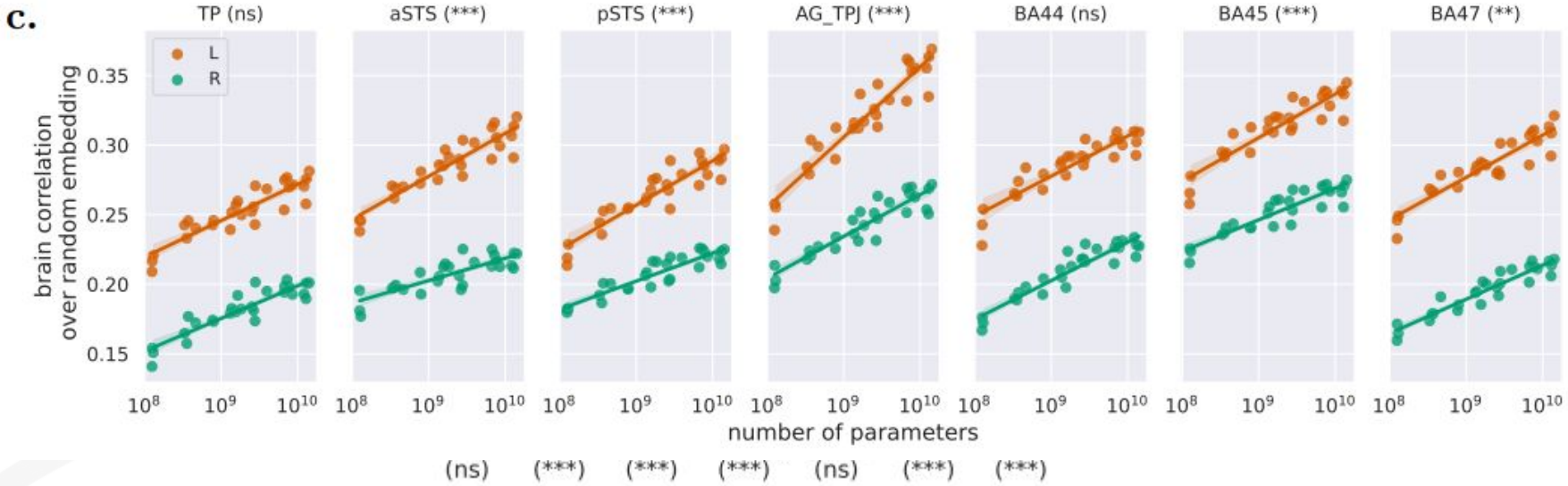
Las capas intermedias son las que dan mayor correlación



Mejores modelos, mayor lateralización en la correlación



Mejores modelos, mayor lateralización en la correlación



Mayor correlación con el cerebro, mayor correlación con comprensión lectora

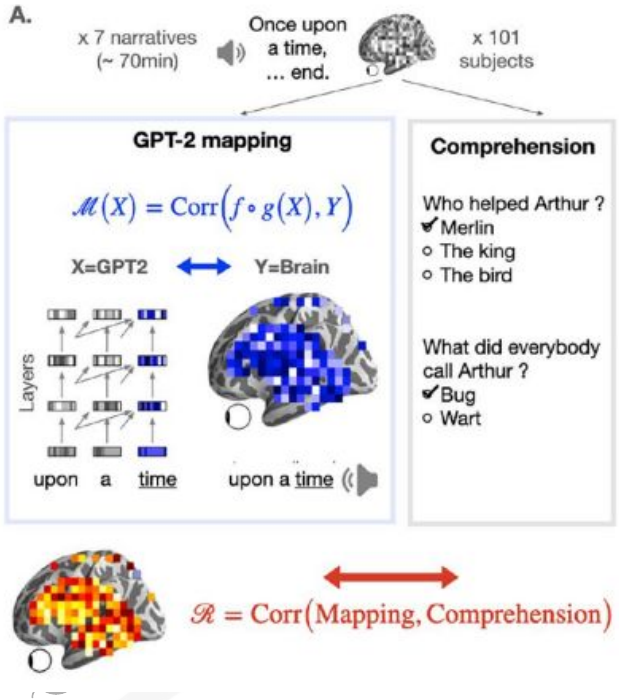
Check for updates

OPEN

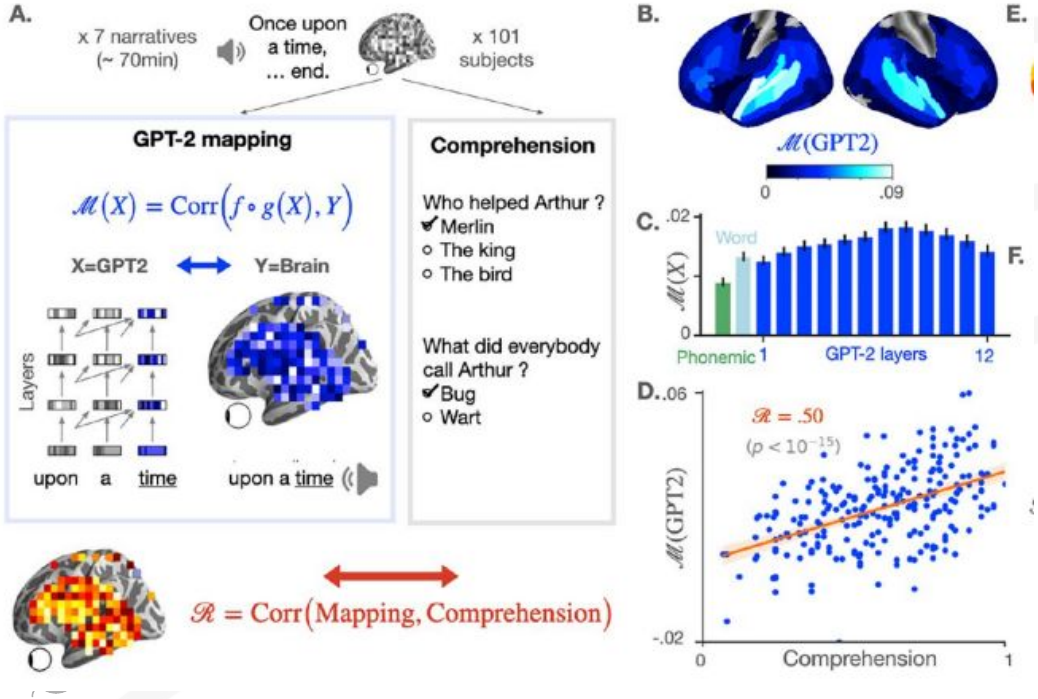
Deep language algorithms predict semantic comprehension from brain activity

Charlotte Caucheteux[1,2]✉, Alexandre Gramfort[2] & Jean-Rémi King[1,3]

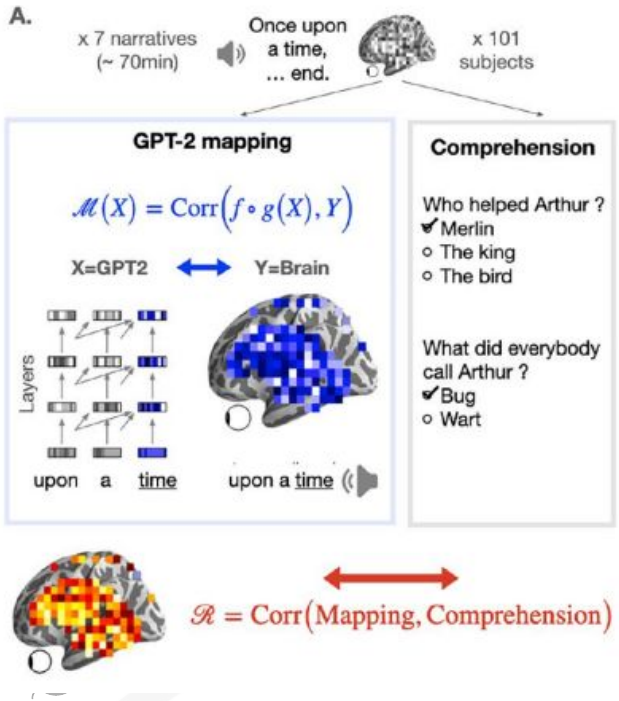
Mayor correlación con el cerebro, mayor correlación con comprensión lectora

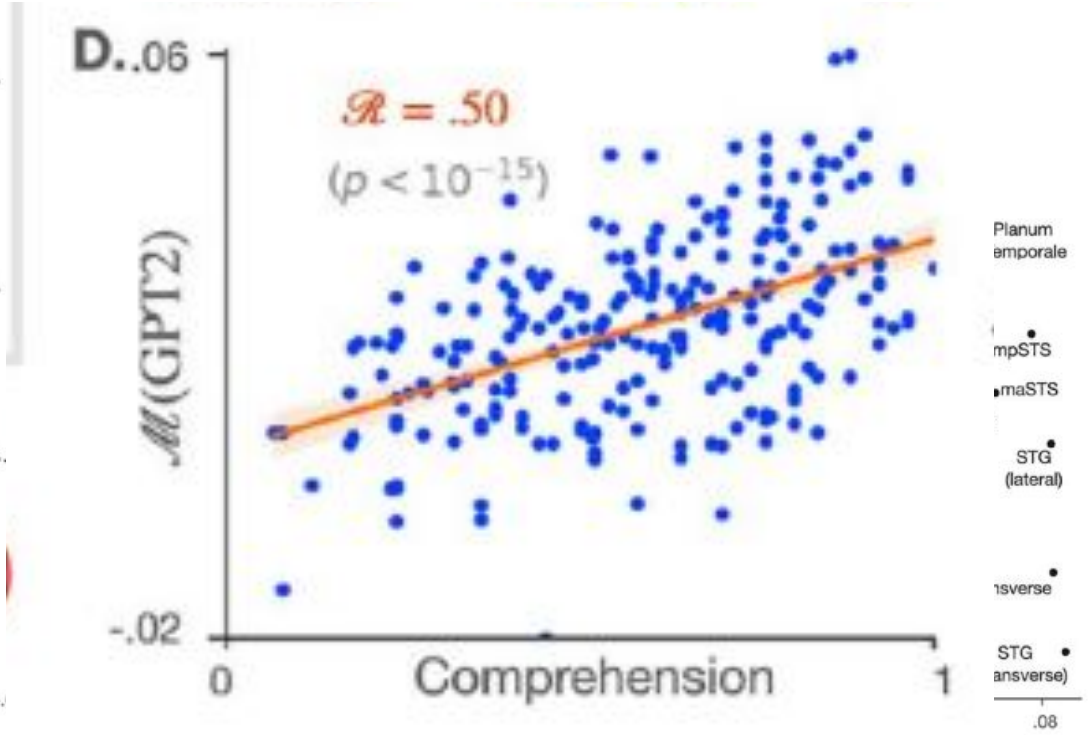


Mayor correlación con el cerebro, mayor correlación con comprensión lectora



Mayor correlación con el cerebro, mayor correlación con comprensión lectora





Entonces

- Los LLMs también permiten estudiar cómo el cerebro procesa el lenguaje
- ¿Por qué? ¿Es que ambos sistemas realizan tareas similares? Evidencia mixta
- Nos ayudan a refinar las preguntas, y también a **cuantificar el lenguaje**
- Esto recién comienza…

Gracias!